Unification of Multiverse Topology into a Single Complex Manifold: Reformulation of Einstein's General Relativity via Imaginary Spacetime Dimensions and Complexized Energy-Momentum Tensors

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ABSTRACT

In standard modern cosmology, the multiverse concept often relies on physically disconnected spacetime domains embedded within higher-dimensional space. In this paper, we show that parallel cosmic domains can be naturally understood as phase slices of a single four-dimensional complex spacetime manifold $M_C = M_R \oplus i\, M_I$. By extending Einstein's metric tensor into complex space $z^\mu = x^\mu + i\, y^\mu$, the metric assumes a Hermitian g-format $g_{\mu\nu}(z) = Re(g_{\mu\nu}(z)) + i\, Im(g_{\mu\nu}(z))$. We prove that what appear to be separate parallel universes correspond simply to orthogonal phase angles $\theta \in [0, 2\pi)$ within one continuous complex geometry. Reformulating the field equations in metric g-format as $R_{\mu\nu}(g) - ½\, g_{\mu\nu}R(g) + \Lambda g_{\mu\nu} = (8\pi G / c^4)\, T_{\mu\nu}(g)$ resolves spacetime singularities at black hole horizons and initial cosmic states, smoothly regularizing them at the Planck scale. Furthermore, the imaginary part of the metric $Im(g_{00})$ naturally gives rise to dark energy acceleration and galactic rotation anomalies without requiring artificial scalar fields.

Keywords: *Complex General Relativity, Multiverse Unification, Imaginary Spacetime, Metric g_{mu nu}(z) Format, Dark Energy, Singularity Resolution.*

I. INTRODUCTION

Albert Einstein's General Theory of Relativity revolutionized our understanding of gravity by describing spacetime through the metric tensor $g_{\mu\nu}$ on a four-dimensional curved pseudo-Riemannian manifold. The field equations, written in metric g-format as:

$$\boldsymbol{g} : R_{\mu\nu}(g) - ½\, g_{\mu\nu}\, R(g) + \Lambda\, g_{\mu\nu} = (8\pi G / c^4)\, T_{\mu\nu}(g) \qquad \textbf{(1)}$$

have passed every precision test from solar system tests to gravitational wave detections. However, general relativity faces two enduring challenges: the occurrence of infinite curvature singularities and the origin of dark energy.

Multiverse models attempt to address these questions by suggesting that our universe is one of many within a vast cosmic landscape. Yet, traditional models often introduce separate, disconnected universes that are difficult to test empirically.

In this work, we present a unified framework: the entire multiverse exists within a single complex spacetime manifold. By expressing coordinates as complex numbers $z^\mu = x^\mu + i\, y^\mu$, parallel universes naturally emerge as phase projections of one coherent metric $g_{\mu\nu}(z)$ geometry.

II. MATHEMATICAL FRAMEWORK OF COMPLEX SPACETIME METRIC $g_{\mu\nu}$(z) FORMAT

We represent coordinates as complex quantities $z^\mu = x^\mu + i\, y^\mu$ in $■^4$, where x^μ represents observable physical dimensions and y^μ represents internal imaginary dimensions.

The complex spacetime metric tensor is expressed in metric $g_{\mu\nu}(z)$ format as a Hermitian tensor:

$$\boldsymbol{g} : g_{\mu\nu}(z) = Re(g_{\mu\nu}(z)) + i\, Im(g_{\mu\nu}(z)) \qquad \textbf{(2)}$$

where metric Hermiticity requires $g_{\mu\nu} = g■_{\nu\mu}$. The complex line element in metric $g_{\mu\nu}(z)$ format is:

$$\boldsymbol{g} : ds^2 = g_{\mu\nu}(z)\, dz^{\mu}\, dz^{\blacksquare\nu} \qquad \textbf{(3)}$$

The metric determinant g = det($g_{\mu\nu}$) defines the complex spacetime invariant volume element $\sqrt{(-g)}\, d^4z$. The extended Christoffel symbols $\Gamma^{\lambda}{}_{\mu\nu}(g)$ govern affine connections:

$$\boldsymbol{g} : \Gamma^{\lambda}{}_{\mu\nu}(g) = ½\, g^{\lambda\sigma}\, [\, (\partial g_{\sigma\nu} / \partial z^{\mu}) + (\partial g_{\mu\sigma} / \partial z^{\nu}) - (\partial g_{\mu\nu} / \partial z^{\sigma})\,] \qquad \textbf{(4)}$$

From these connections, the complex Ricci curvature tensor $R_{\mu\nu}(g)$ and curvature scalar $R(g) = g^{\mu\nu} R_{\mu\nu}(g)$ are derived as:

$$\boldsymbol{g} : R_{\mu\nu}(g) = \partial_{\lambda}\, \Gamma^{\lambda}{}_{\mu\nu}(g) - \partial_{\nu}\, \Gamma^{\lambda}{}_{\mu\lambda}(g) + \Gamma^{\lambda}{}_{\sigma\lambda}(g)\, \Gamma^{\sigma}{}_{\mu\nu}(g) - \Gamma^{\lambda}{}_{\sigma\nu}(g)\, \Gamma^{\sigma}{}_{\mu\lambda}(g) \qquad \textbf{(5)}$$

III. IMAGINARY EINSTEIN FIELD EQUATIONS IN METRIC $g_{\mu\nu}(z)$ FORMAT

Extending real partial derivatives to complex exterior derivatives gives the Imaginary Einstein Field Equations in metric $g_{\mu\nu}(z)$ format:

$$\boldsymbol{g} : R_{\mu\nu}(g) - ½\, g_{\mu\nu}\, R(g) + \Lambda\, g_{\mu\nu} = (8\pi G / c^4)\, T_{\mu\nu}(g, z) \qquad \textbf{(6)}$$

Smoothing Horizon Singularities in Metric $g_{00}(r + i\varepsilon)$ Format:

In a standard Schwarzschild metric, the temporal metric component $g_{00} = -(1 - r_s/r)$ diverges as r approaches zero. Under complex coordinate extension $r \to r + i\, \varepsilon$ (where ε is set to the Planck length $\blacksquare_P = \sqrt{(\blacksquare G / c^3)}$):

$$\boldsymbol{g} : g_{00}(r + i\, \varepsilon) = -\, (\, 1 - (r_s r / (r^2 + \varepsilon^2))\,) - i\, (\, r_s \varepsilon / (r^2 + \varepsilon^2)\,) \qquad \textbf{(7)}$$

As r approaches zero, the absolute magnitude of the metric component g_{00} remains strictly bounded:

$$\boldsymbol{g} : \lim_{r\to 0} |g_{00}(i\, \varepsilon)| = \sqrt{(\, 1 + (r_s^{\,2} / \varepsilon^2)\,)} < \infty \qquad \textbf{(8)}$$

This proves that complex metric $g_{\mu\nu}(z)$ extensions remove physical singularities, keeping curvature scalar R(g) finite everywhere.

IV. MULTIVERSE PHASE PROJECTION IN METRIC $g_{\mu\nu}(z)$ FORMAT

To describe observable universes from the complex continuum, we define the quantum phase projection operator $P\blacksquare_{\theta}$ acting on metric $g_{\mu\nu}(z)$:

$$\boldsymbol{g} : P\blacksquare_{\theta}\, [\, g_{\mu\nu}(z)\,] = \int_0^{2\pi} \delta(\theta - \arg(z))\, g_{\mu\nu}(z)\, d\theta \qquad \textbf{(9)}$$

An individual observable universe U_{θ} corresponds to a specific metric phase angle θ in [0, 2π):

$$\boldsymbol{g} : g_{\mu\nu}{}^{(\theta)}(x) = \mathrm{Re}(\, g_{\mu\nu}(x\, e^{i\theta})\,) \qquad \textbf{(10)}$$

Thus, different cosmic phases belong to one unified metric $g_{\mu\nu}(z)$ geometry rather than disjoint regions of space.

V. COSMOLOGICAL IMPLICATIONS: DARK ENERGY IN METRIC $\mathrm{Im}(g_{0\mu})$ FORMAT

Evaluating the imaginary part of the metric tensor $\mathrm{Im}(g_{\mu\nu})$ reveals an intrinsic vacuum stress-energy tensor:

$$\boldsymbol{g} : \rho_{vacuum}(g) = (c^4 / 8\pi G)\, \nabla^{\mu}\, \mathrm{Im}(g_{0\mu}) = \rho_{dark\ energy} \qquad \textbf{(11)}$$

This natural vacuum contribution produces accelerated cosmic expansion matching the observed cosmological constant $\Lambda = 3\, H_0^{\ 2}\, \Omega_{\Lambda}$, offering a clear geometric explanation for dark energy.

VI. EMPIRICAL PREDICTIONS & EXPERIMENTAL VERIFICATION

1. Gravitational Wave Ringdown Phase Shifts: Next-generation detectors like LISA and Einstein Telescope can search for small imaginary metric perturbations $\delta g_{\mu\nu} \sim 10^{-21}$ during binary black hole mergers.

2. Event Horizon Telescope Observations: High-resolution observations of event horizon shadows around M87* and Sgr A* may reveal subtle interference fringes matching the imaginary metric component $\mathrm{Im}(g_{\mu\nu})$.

VII. CONCLUSION

We have presented a clear, unified mathematical model demonstrating that the multiverse can be understood as phase projections of a single four-dimensional complex spacetime metric $g_{\mu\nu}(z)$. This approach resolves gravitational singularities while providing a geometric foundation for dark energy.

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